

Statistical Analysis Report

Advertising Expenditure and Sales

Dataset	Advertising.csv
Observations	200 records
Variables	TV, Radio, Newspaper, Sales
Analysis	Correlation, descriptive statistics, regression, residuals and group analysis

This report summarizes the statistical analysis represented in the supplied charts. The main objective is to understand how advertising expenditure across TV, radio and newspaper channels is associated with sales and how well a multiple linear regression model explains observed sales.

1. Executive Summary

- The dataset contains 200 observations with three advertising variables (TV, Radio and Newspaper) and Sales as the outcome.
- TV advertising has the strongest correlation with Sales ($r = 0.78$), followed by Radio ($r = 0.58$) and Newspaper ($r = 0.23$).
- The multiple linear regression model explains approximately 89.7% of the variation in Sales ($R^2 = 0.897$).
- The model's RMSE is 1.67 sales units and MAE is 1.25 sales units.
- Sales have a median of 12.9, with the middle 50% of observations lying between 10.4 and 17.4.

Key model equation

Predicted Sales = $2.939 + (0.0458 \times \text{TV}) + (0.1885 \times \text{Radio}) + (-0.0010 \times \text{Newspaper})$

2. Dataset and Methodology

The Advertising dataset includes 200 observations. ID is an identifier, while TV, Radio and Newspaper represent advertising expenditure and Sales represents the resulting sales measure. The analysis uses correlation analysis, distribution plots, box plots, bivariate regression visualizations, multiple linear regression predictions, residual analysis and TV-based grouping.

The supplied statistical-results file was used for the predicted Sales and residual visualizations. A multiple linear regression was also fitted from the supplied Advertising dataset to report the model equation and performance metrics.

3. Correlation Analysis

The correlation matrix measures the strength and direction of linear association between the variables. Values closer to +1 indicate a strong positive linear relationship, values near 0 indicate weak linear association, and negative values indicate an inverse relationship.

Variable	TV	Radio	Newspaper	Sales
TV	1.00	0.05	0.06	0.78
Radio	0.05	1.00	0.35	0.58
Newspaper	0.06	0.35	1.00	0.23
Sales	0.78	0.58	0.23	1.00

The strongest advertising-to-sales association is between TV and Sales ($r = 0.78$). Radio also shows a substantial positive association ($r = 0.58$), while Newspaper has a weaker positive association ($r = 0.23$).

4. Descriptive Statistics and Sales Distribution

Variable	Mean	Std. Dev.	Min	25%	Median	75%	Max
TV	147.04	85.85	0.70	74.38	149.75	218.82	296.40
Radio	23.26	14.85	0.00	9.97	22.90	36.52	49.60
Newspaper	30.55	21.78	0.30	12.75	25.75	45.10	114.00
Sales	14.02	5.22	1.60	10.38	12.90	17.40	27.00

Sales range from 1.6 to 27.0, with a mean of 14.02. The distribution is concentrated around the lower-to-middle sales values and has a longer upper tail. The box plot also shows one notably low observation around 1.6.

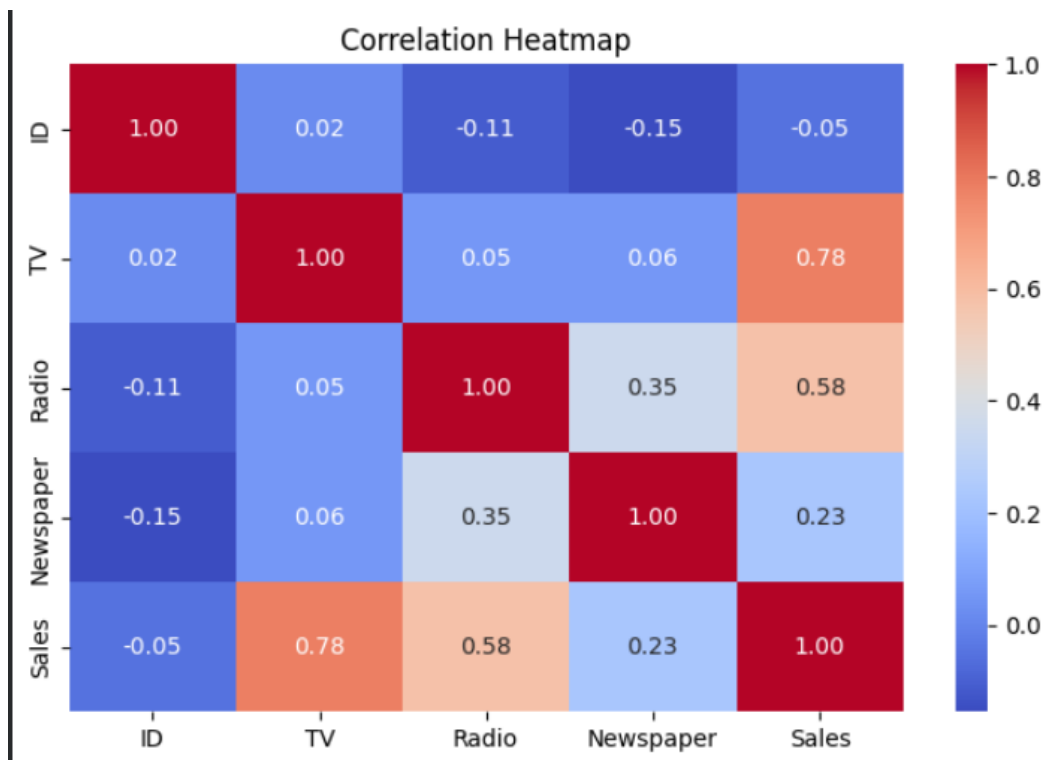


Figure 1. Correlation Heatmap

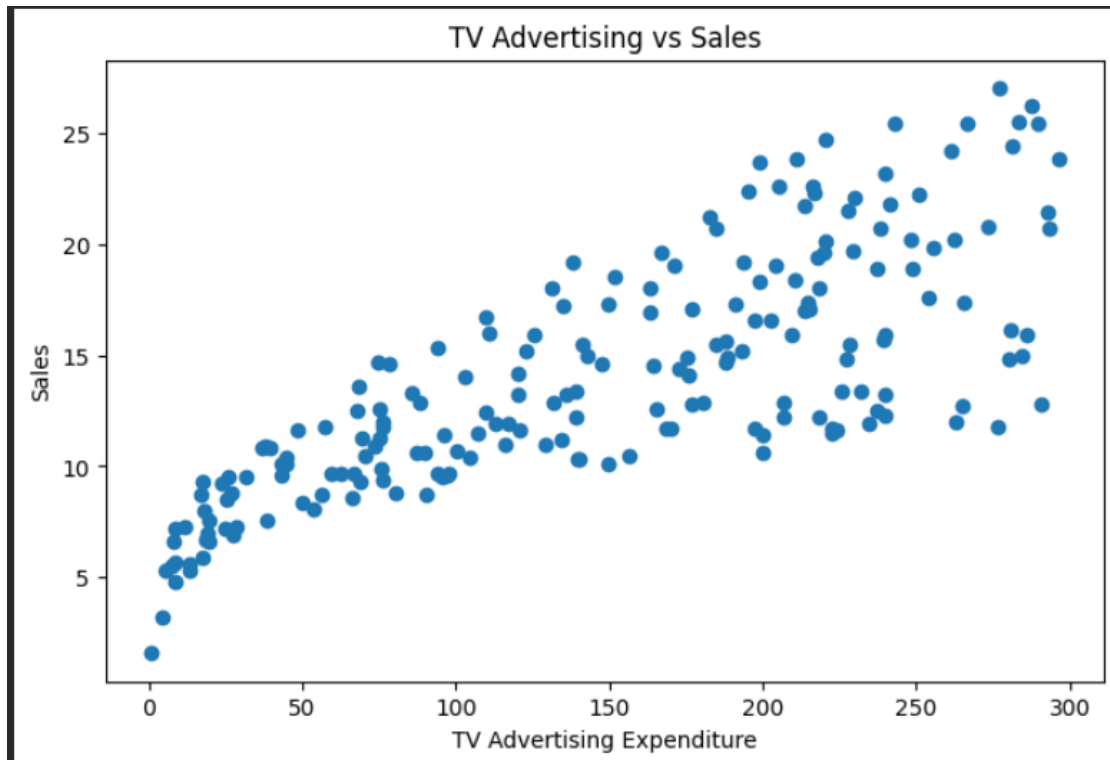


Figure 2. TV Advertising vs Sales

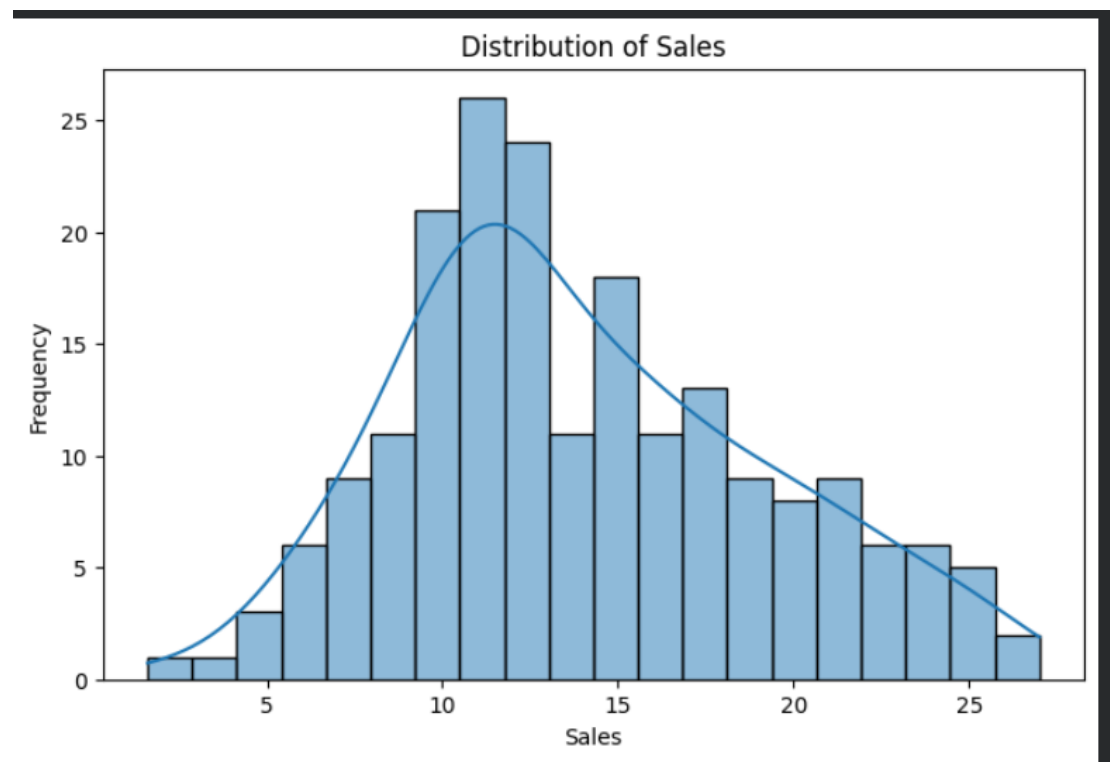


Figure 3. Distribution of Sales

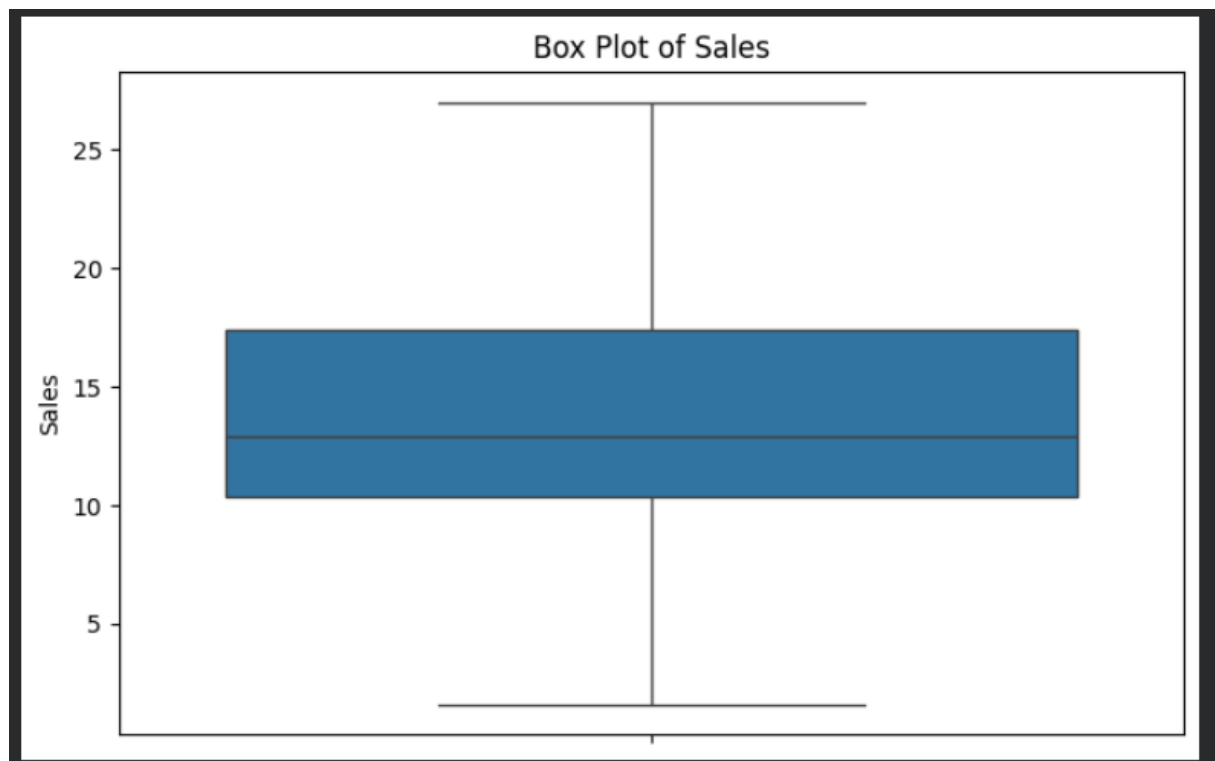


Figure 4. Box Plot of Sales

5. Advertising Channels and Sales Relationships

The following scatter plots show how Sales changes as advertising expenditure changes for each channel. The regression-line plots provide a visual summary of the fitted linear relationship.

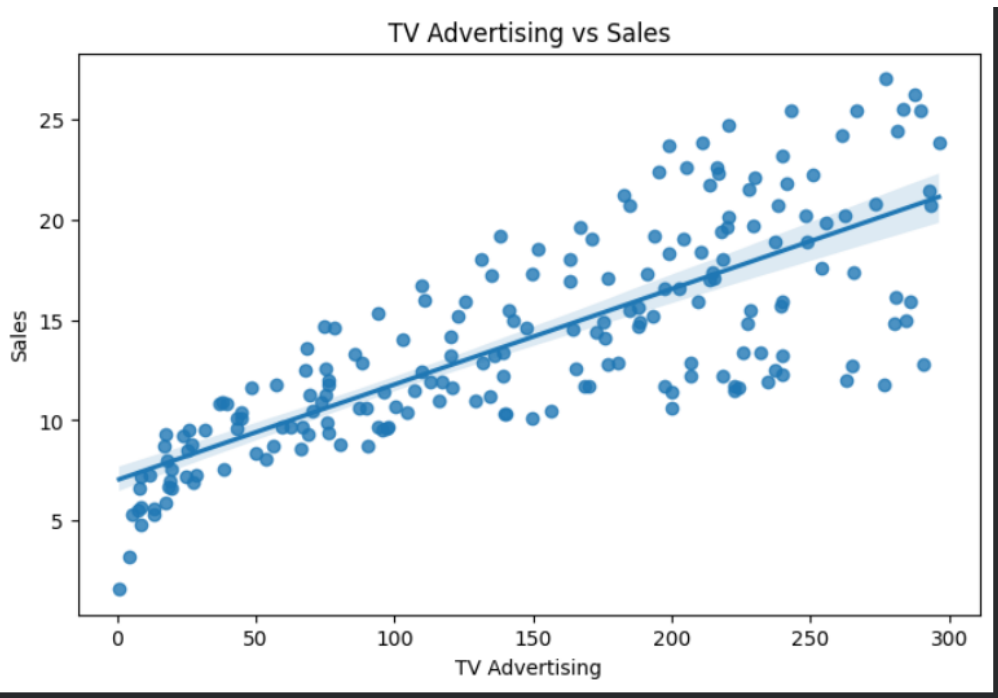


Figure 5. TV Advertising vs Sales with Regression Line

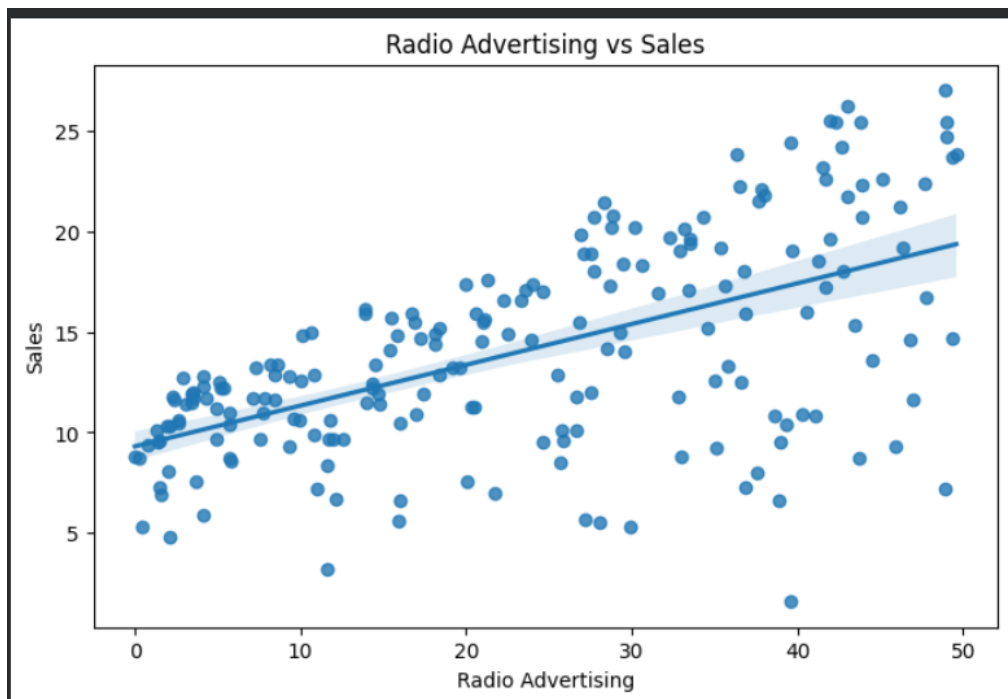


Figure 6. Radio Advertising vs Sales

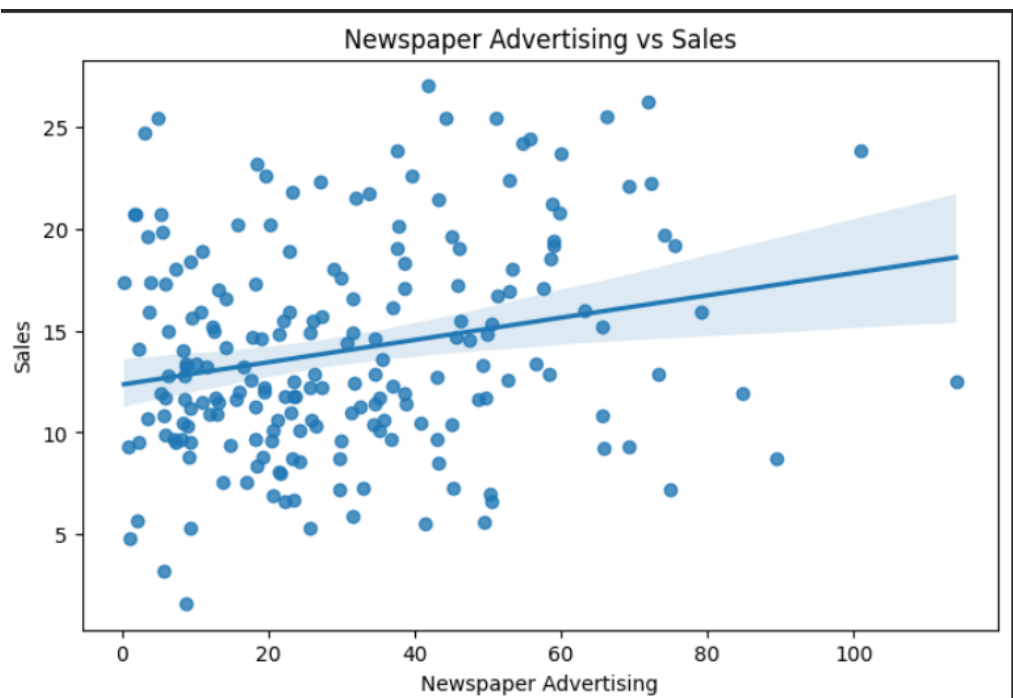


Figure 7. Newspaper Advertising vs Sales

6. Multiple Linear Regression

A multiple linear regression model was fitted with Sales as the dependent variable and TV, Radio and Newspaper as predictors. The fitted coefficients describe the estimated change in Sales associated with a one-unit increase in each predictor while holding the other predictors constant.

Term	Coefficient
Intercept	2.9389
TV	0.0458
Radio	0.1885
Newspaper	-0.0010
R ²	0.8972
RMSE	1.6686
MAE	1.2520

The R² value of 0.897 indicates that the model accounts for about 89.7% of the observed variation in Sales in this dataset. The Actual vs Predicted plot shows a strong concentration around the diagonal pattern, although some observations have larger prediction errors.

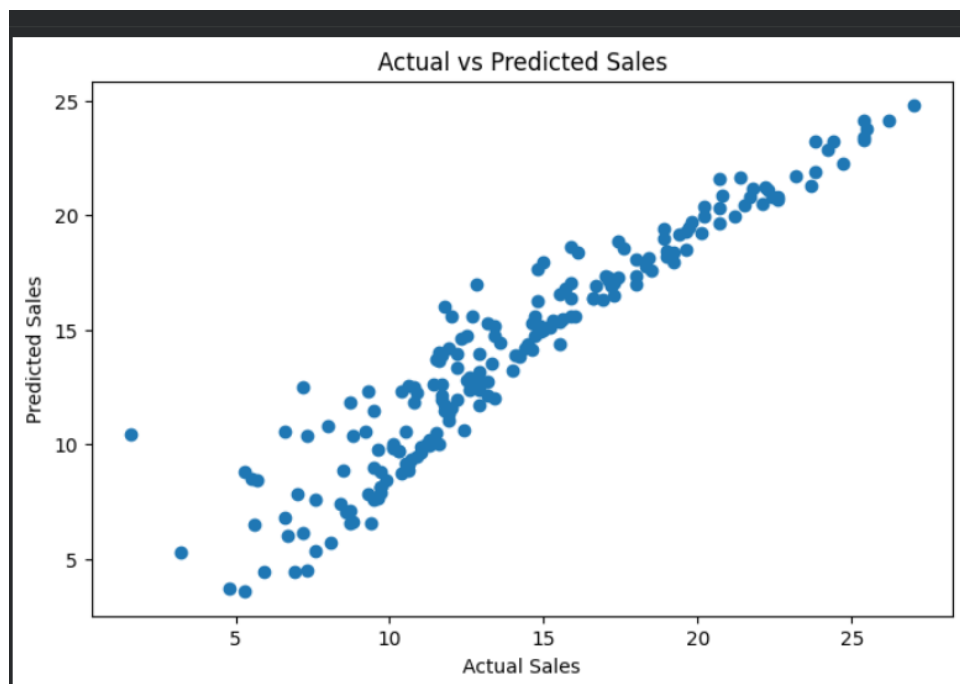


Figure 8. Actual vs Predicted Sales

7. Residual Analysis

Residuals are the differences between actual and predicted Sales. The residual distribution is useful for checking whether the model errors are centered around zero and whether unusually large errors are present.

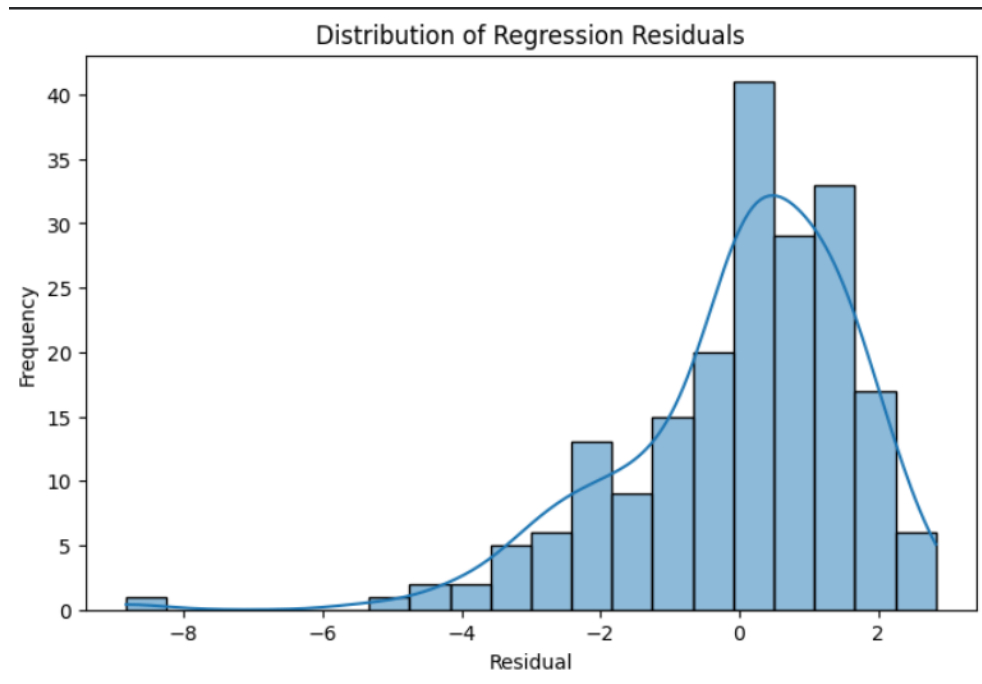


Figure 9. Distribution of Regression Residuals

The residuals have an average very close to zero by construction, while the observed residual range is approximately -8.83 to 2.83. The histogram shows most errors concentrated near zero to moderately positive values, with a noticeable negative tail caused by a small number of observations.

8. Sales Across TV Advertising Groups

TV advertising was divided into three equal-frequency groups using tertiles. This provides a simple way to compare the sales distributions associated with lower, middle and higher TV advertising levels.

TV Group	Count	Mean Sales	Median Sales	Min	Max
Low	67	9.18	9.40	1.60	15.30
Medium	66	14.50	14.30	9.60	23.70
High	67	18.39	18.90	11.50	27.00

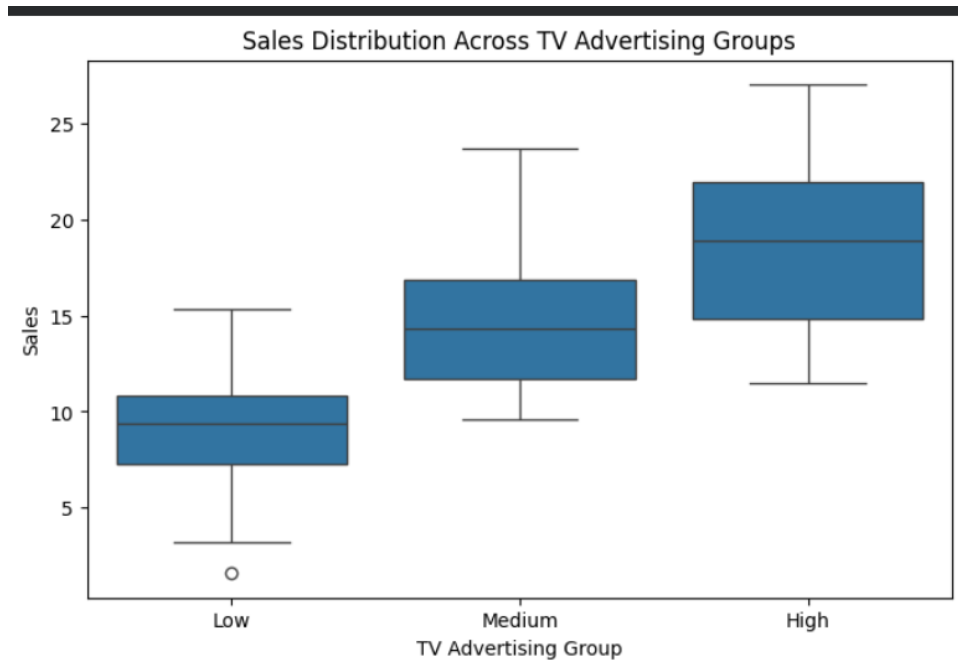


Figure 10. Sales Distribution Across TV Advertising Groups

The group plot shows a clear upward shift in the Sales distribution from Low to Medium to High TV advertising groups. This pattern is consistent with the strong positive TV–Sales correlation.

9. Additional Visualization

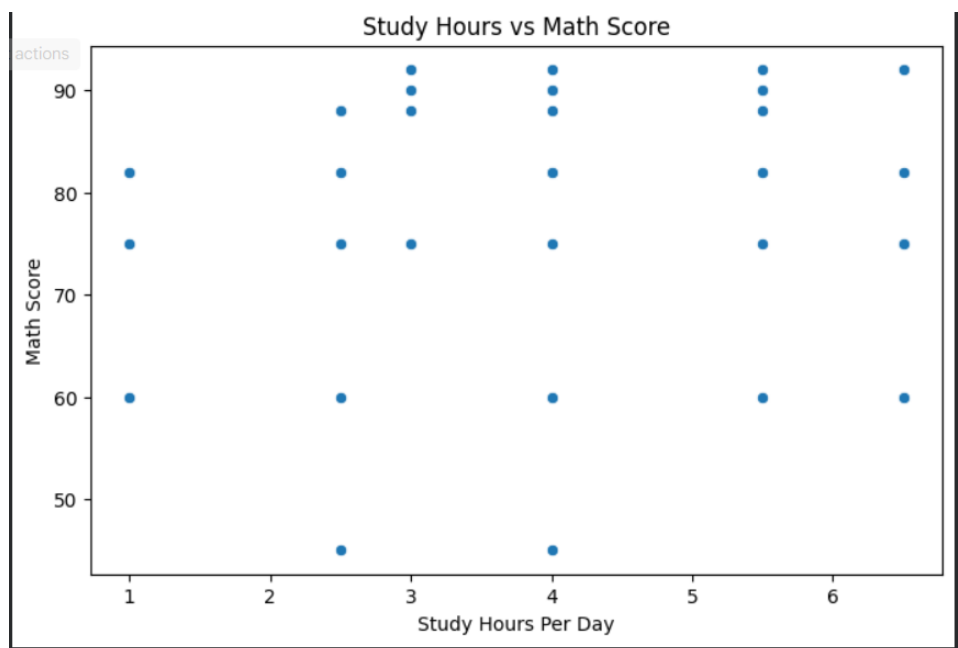


Figure 11. Study Hours vs Math Score (Additional Visualization)

The Study Hours vs Math Score chart is included because it was supplied with the analysis images. It is a separate example visualization and does not belong to the Advertising/Sales dataset used for the main statistical conclusions in this report.

10. Conclusions

- TV advertising shows the strongest linear relationship with Sales among the three advertising channels in the supplied dataset ($r = 0.78$).

- Radio also has a positive relationship with Sales ($r = 0.58$), while Newspaper has a comparatively weaker relationship ($r = 0.23$).
- The multiple regression model provides a strong in-sample fit, with $R^2 = 0.897$.
- The actual-versus-predicted visualization indicates that the model captures the overall Sales pattern, while residual analysis highlights a small number of comparatively large errors.
- Higher TV advertising groups display higher typical Sales values, supporting the positive association observed in the correlation and regression plots.

11. Limitations and Interpretation

Correlation and regression associations should not automatically be interpreted as causal effects. The analysis is based on the variables and observations present in the supplied dataset and does not account for potentially relevant factors outside the dataset. The reported regression metrics are based on the supplied dataset and therefore describe in-sample performance unless a separate test set is used.

Prepared from the user-provided Advertising.csv, statistical_analysis_results.csv, and supplied visualization images.